

SAGONA ISLAND, NL, Canada

WMO: 714080

Lat: 47.367N

Lon: 55.795W

Elev: 60

StdP: 100.61

Time Zone: -3.50 (NAF)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.7	-10.1	-16.7	0.9	-11.2	-15.1	1.0	-9.7	27.8	1.2	25.8	-1.0	12.0	310	1.016

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.8	19.4	17.7	18.6	17.2	17.9	16.6	18.5	19.2	17.8	18.5	17.1	17.8	6.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.2	18.8	17.5	12.6	18.2	16.8	12.0	17.6	52.7	19.2	50.6	18.6	48.4	17.8	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	21.1	18.8	-13.9	22.4	2.4	2.1	-15.7	23.9	-17.1	25.1	-18.5	26.2	-20.2	27.7
			-14.4	20.6	2.7	2.0	-16.3	22.1	-17.9	23.2	-19.4	24.4	-21.4	25.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-2.8	-3.2	-1.4	2.0	5.9	9.6	14.0	16.2	13.4	8.9	4.6	0.2
	DBStd	7.09	3.51	3.74	3.21	2.32	2.17	2.20	1.98	1.58	2.31	2.69	3.12	3.50
	HDD10.0	2046	396	371	355	240	129	33	0	0	3	53	162	304
	HDD18.3	4625	654	604	613	490	387	262	133	66	149	293	411	563
	CDD10.0	465	0	0	0	0	1	21	126	194	104	19	1	0
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	1	0	0	0	0	0	1	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	9.2	11.9	11.2	10.2	9.0	7.6	6.8	6.5	7.1	8.4	9.8	10.6	11.6
Precipitation	PrecAvg	1660	151	142	131	125	122	106	123	104	155	175	174	160
	PrecMax	2060	257	249	224	277	174	183	222	149	235	356	229	259
	PrecMin	1455	89	76	74	51	33	43	56	70	80	95	86	109
	PrecStd	147	41	43	47	61	35	38	44	22	44	65	35	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	6.8	6.2	9.7	13.8	17.7	20.5	20.9	19.6	16.2	12.6	9.4
		MCWB	7.1	6.5	4.4	7.9	10.1	14.1	18.3	18.5	17.0	15.2	12.3	9.1
	2%	DB	4.9	4.7	4.5	7.7	11.5	15.1	18.6	19.8	18.2	14.7	11.4	7.6
		MCWB	4.7	4.3	3.1	6.4	9.1	13.3	17.0	17.9	16.9	14.0	11.1	7.3
	5%	DB	3.3	3.0	3.4	6.2	10.1	13.9	17.4	19.1	17.4	13.6	10.4	6.4
		MCWB	2.9	2.6	2.2	5.0	8.2	12.3	16.1	17.6	16.2	12.7	10.1	6.1
	10%	DB	2.0	1.5	2.3	5.0	8.9	12.7	16.5	18.4	16.5	12.6	9.1	5.0
		MCWB	1.5	1.0	1.3	4.0	7.3	11.5	15.5	17.1	15.3	11.6	8.6	4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	6.6	5.1	8.5	11.3	15.6	19.7	19.7	18.4	16.2	12.6	9.9
		MCDB	7.6	6.8	5.6	9.0	12.7	16.7	21.2	20.4	18.8	16.4	12.8	10.0
	2%	WB	4.9	4.5	3.1	6.8	9.9	14.0	18.0	18.7	17.5	14.7	11.3	7.7
		MCDB	5.0	4.7	3.8	7.2	11.2	15.0	18.9	19.3	17.9	15.0	11.4	7.9
	5%	WB	3.1	2.3	2.1	5.4	8.9	12.8	16.9	18.2	16.6	13.3	10.3	6.4
		MCDB	3.3	2.5	2.7	6.1	10.1	13.5	17.6	18.9	17.2	13.7	10.5	6.6
	10%	WB	1.7	0.8	1.1	4.3	7.8	11.7	16.1	17.6	15.7	12.3	9.0	4.6
		MCDB	2.0	1.1	1.7	5.0	9.1	12.4	16.7	18.3	16.3	12.8	9.2	4.9
Mean Daily Temperature Range	5% DB	MDBR	4.8	4.7	4.5	4.4	4.9	4.6	3.9	3.8	4.0	3.7	4.0	4.3
		MCDBR	6.7	6.3	5.3	6.1	7.1	6.5	5.3	4.5	4.6	4.4	5.3	5.7
		MCWBR	7.5	7.4	5.5	5.1	5.2	5.2	4.1	3.6	4.2	4.6	6.1	7.0
	5% WB	MCDBR	6.8	7.0	5.5	6.1	7.0	6.4	5.2	4.2	4.3	4.5	5.5	6.3
		MCWBR	7.5	7.4	5.6	5.4	5.4	5.3	4.3	3.6	4.1	4.9	6.2	7.1
Clear-Sky Solar Irradiance	taub	0.317	0.314	0.330	0.370	0.375	0.399	0.424	0.404	0.369	0.345	0.320	0.308	
	taud	2.345	2.436	2.435	2.347	2.397	2.360	2.336	2.389	2.475	2.494	2.491	2.402	
	Ebn at Noon	754	854	893	883	889	866	837	840	842	808	758	724	
	Edh at Noon	73	82	96	115	114	119	121	110	92	78	64	62	
All-Sky Solar Radiation	RadAvg	1.04	1.65	2.81	4.18	5.03	5.14	4.88	4.36	3.52	2.20	1.21	0.85	
	RadStd	0.08	0.18	0.20	0.37	0.39	0.53	0.38	0.26	0.22	0.16	0.14	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	+0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page